Diamond clause when both modalities are primitive: derived via duality and the box clause

The Open Logic Project contributors

openlogic:normal-modal-logic/completeness:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: claim_key carries @1321 (second instance of this label; see the sibling card at 1256). This branch cites prop:box, whose text sits in the Box-primitive branch of the same source - both spans assemble verbatim.

Editorial note. The label prop:diamond names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Editorial note. OLP reuses this label inside the unit; the card is identified by its statement span (source lines 1321-1326) and the claim key carries the statement start line.

Allowed standard prerequisites (editorial).

- Negation clause of maximal consistency. Same part, \ollabel{prop:ccs-properties} contracted to item \ollabel{prop:ccs-lnot} (line 504 region). Lines 1326 and 1335: membership of a negation as non-membership, twice.
- Box clause of the canonical model. Same section, \ollabel{prop:box} (source line 1185, own card). Line 1328: rewrites the failure of Box not-A at Gamma into a witnessing Delta under the inverse-box inclusion.
- Accessibility duality lemma. Same section, \ollabel{lem:box-iff-diamond} (source line 1247, own card). Line 1331: converts the inverse-box inclusion on Delta into the Diamond-image inclusion demanded by the statement.
- Duality axiom pattern and closure. Normal modal logic: Diamond A defined/equivalent to not Box not-A, cited via the Dual macro. Line 1325: first equivalence of the display-free chain.

Source reference. Original lemma lem-box-iff-diamond: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1291–1291. Prerequisite pointer invoked at proof line(s) 1337; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-box: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1229–1229. Prerequisite pointer invoked at proof line(s) 1333; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-lnot: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 530–530. Prerequisite pointer invoked at proof line(s) 1332; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-properties: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 519–519. Prerequisite pointer invoked at proof line(s) 1332; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *If Γ is complete Σ-consistent, then $\Diamond A \in \Gamma$ if and only if for some complete Σ-consistent Δ such that $\Diamond\Delta \subseteq \Gamma$, it holds that $A \in \Delta$.*

Proof. Suppose Γ is complete Σ-consistent. $\Diamond A \in \Gamma$ iff $\neg\Box\neg A \in \Gamma$ by d and closure. $\neg\Box\neg A \in \Gamma$ iff $\Box\neg A \notin \Gamma$ by prop-ccs-propertiesprop-ccs-lnot since Γ is complete Σ-consistent. By prop-box, $\Box\neg A \notin \Gamma$ iff, for some complete Σ-consistent Δ with $\Box^{-1}\Gamma \subseteq \Delta$, $\neg A \notin \Delta$. Now consider any such Δ. By lem-box-iff-diamond, $\Box^{-1}\Gamma \subseteq \Delta$ iff $\Diamond\Delta \subseteq \Gamma$. Also, $\neg A \notin \Delta$ iff $A \in \Delta$ by prop-ccs-propertiesprop-ccs-lnot. So $\Diamond A \in \Gamma$ iff, for some complete Σ-consistent Δ with $\Diamond\Delta \subseteq \Gamma$, $A \in \Delta$. $\square$